

Keeping up with the AIs

Mistral's trillion-parameter open model, and running AI on your own laptop

Issue #8

Wed, 07 Oct 2026

 ~8 min read

⚡ If you only have one minute

1. **Mistral unveiled Large 4, nicknamed "Le Chonk":** a ~1-trillion-parameter model you can try by API today. The weights are due 27 October.
2. **OpenAI published 722 maths papers written by an unreleased model**, many with machine-checked proofs. Anthropic opened its strongest cyber models to more security teams.
3. **Part 1 explains open-weight vs closed models**, and how to run one on your own laptop in 10 minutes.

Today in three numbers

49B

parameters **active per word** in Mistral's ~1-trillion-parameter Large 4

722

maths **manuscripts** OpenAI released from an unreleased model

129,000

verified **software bugs** found with Anthropic's cyber models, April–July

PART 1

Learn one thing

Open-weight vs closed models (and running AI at home)

4 MIN READ

MODULE 1 · HOW LLMS WORK

Every model is, in the end, a giant file of numbers called **weights**. The real question is whether you're allowed to download that file.

Closed models (GPT, Claude, Gemini) stay on the company's servers. You send text over the internet and **pay per token**. **Open-weight models** (Qwen, DeepSeek, Mistral, Reflection's Beam) let you **download the weights** and run them on your own machine or server.

Analogy: a closed model is a **restaurant**. The food is great, but you eat there, at their prices, by their rules. An open-weight model is a **frozen meal you take home**: you can reheat it anywhere and add your own spices (fine-tuning). But you usually **don't get the recipe**, meaning the training data and code. That's why "**open-weight**" is not the same as "**open source**."

Closed model (API)

- Always the newest, strongest models
- No hardware to buy; pay per token
- Your data goes to their servers
- They can change or retire the model

VS

Open-weight model

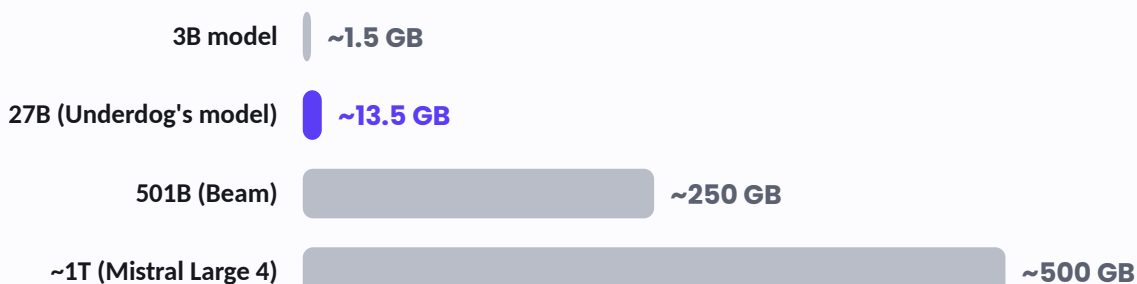
- Runs on your laptop or server
- Data never leaves your machine
- No per-token bill, but you pay for hardware
- Check the licence: they vary a lot

Licences differ: Reflection plans Apache 2.0 for Beam; VentureBeat reports a custom Mistral licence for Large 4.

WILL IT FIT ON MY LAPTOP?

The weights must fit in memory. A handy rule: $\text{memory} \approx \text{parameters} \times \text{bits per weight} \div 8$. Squeezing each weight from 16 bits down to about 4 is called **quantization**, and it's how big models fit on normal machines.

Rough memory just for the weights, at ~4 bits per weight



Our own arithmetic using the rule above; real files and running overhead add more. LM Studio recommends 16 GB+ of RAM.

So a **3B or 8B model runs on most laptops**, a 27B model needs a well-stocked Mac or a good GPU, and the trillion-parameter giants are **server-only**.

Which should you pick? A simple rule: **prototype on a closed API**, because it's the fastest way to find out if your idea works. Move to open weights when **privacy, offline use or a huge volume of calls** matters more than having the very best model. Many teams end up using both: a small local model for easy, private jobs, and a frontier API for the hard ones.

Why now? In two days, **two big open-weight models** (Beam and Mistral Large 4) were announced, and a startup launched **Underdog, an assistant that runs a 27B model entirely on your PC** (Quick hits).

Jargon buster

- **Weights:** the billions of numbers a model learns in training. They *are* the model.
- **Quantization:** storing each weight with fewer bits, so the model gets smaller and faster, with a small loss in quality.
- **GGUF:** a popular file format for local models; it packs the weights plus the info needed to run them.
(Hugging Face)
- **Licence:** the rules for using the weights. Some allow anything; others limit commercial use.

Try it in 10 minutes

- **No code:** install **LM Studio**, search for a small model (3B–8B), download it and chat. **Then turn off your Wi-Fi.** It keeps working, because nothing leaves your machine. (LM Studio)
- **Some code:** with **Ollama** installed, one line pulls a model from Hugging Face and starts a chat: (Hugging Face)

```
ollama run hf.co/bartowski/Llama-3.2-3B-Instruct-GGUF
```

For devs: swap the cloud for localhost

Ollama speaks the **OpenAI API format**, so existing code mostly works by changing one line: (Ollama)

```
from openai import OpenAI
client = OpenAI(base_url="http://localhost:11434/v1/",
                api_key="ollama") # required but ignored
r = client.chat.completions.create(
    model="hf.co/bartowski/Llama-3.2-3B-Instruct-GGUF",
    messages=[{"role": "user", "content": "Hi!"}])
print(r.choices[0].message.content)
```

- **Pick a quantization** with a tag, like `:Q8_0` or `:IQ3_M`. With no tag, Ollama uses **Q4_K_M**, a good default.
- **Mixture-of-experts models:** the *total* parameters decide the memory you need; the *active* ones decide the speed.

Gotcha: the local API isn't a perfect copy. Ollama's docs say **image URLs**, `tool_choice` and `logprobs` **aren't supported**; send images as base64.

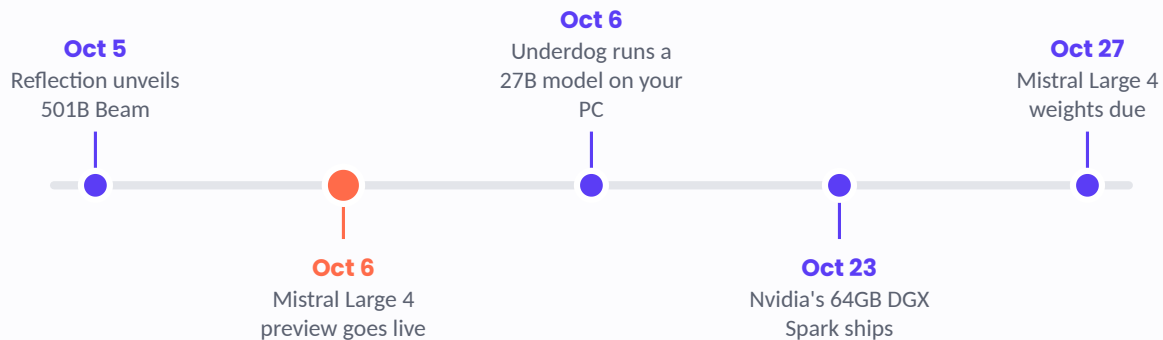
GO DEEPER

- **GGUF** (Hugging Face). The local-model file format and its quantization types.
- **Use Ollama with any GGUF model** (Hugging Face). One-line runs and picking a quantization.
- **OpenAI compatibility** (Ollama). Point existing OpenAI code at your own machine.
- **System requirements** (LM Studio). What Mac, Windows or Linux hardware you need.

What moved in AI

7 min read · Covering Tuesday. October is turning into open-weight month, OpenAI dropped a mountain of maths, and Anthropic's hacker-grade models got a front door.

Open-weight October so far



From TechCrunch (Issue #7 and today), VentureBeat and Tom's Hardware (Issue #4). Beam's weights are due 'later this month', no exact date.

Top story 1 · Mistral's "Le Chonk": a trillion parameters, weights coming

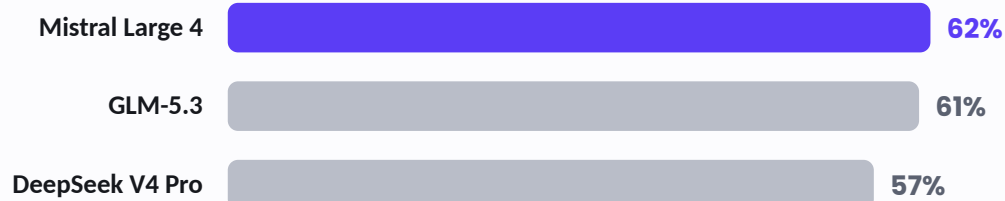
BIG DEAL

OPEN WEIGHTS

DEVS · API PREVIEW

What happened. French lab **Mistral** released **Mistral Large 4** as a public preview on its API. It's a **mixture-of-experts model with about 1 trillion parameters, of which 49 billion work on each token.** It **reads text and images, writes text,** and covers 160+ languages. **Weights are due 27 October.** (VentureBeat) (SiliconANGLE)

DeepSWE v1.1 (agentic coding): Mistral's own numbers



Scores reported by Mistral, via VentureBeat. Not independently verified.

The details that matter:

- **Context: 1 million tokens**, per MarkTechPost, which lists preview pricing of **\$1.36 in / \$4.18 out per million tokens.** (MarkTechPost)
- **Trained on ~3,800 Nvidia Grace Blackwell chips** in Mistral's European data centres. Mistral says **bigger versions** are coming in the next few months. (SiliconANGLE) (MarkTechPost)
- **Money behind it:** Mistral raised a **€3 billion Series D at a €21 billion valuation** in September and says it serves 125+ enterprises, including Airbus and HSBC. (VentureBeat)
- **Aimed at work tasks:** coding, cybersecurity, finance and reading technical drawings and satellite images. **Licence:** VentureBeat reports a custom Mistral licence, so **read it before shipping.** (VentureBeat)

Why it matters. Mistral now offers Europe's answer to the big Chinese open models. With Reflection's Beam on Monday, **companies wary of Chinese weights suddenly have two large options.**

WHO SHOULD CARE



Everyone ● ● ● ● ●



Vibe coders ● ● ● ● ●



Developers ● ● ● ● ●



Hype level ● ● ● ● ●

👁️ **Should you care?** You won't run this on a laptop (~500 GB at 4 bits; see Part 1). **Try it through the API now**, and if you host models for a company, plan your GPUs for late October.

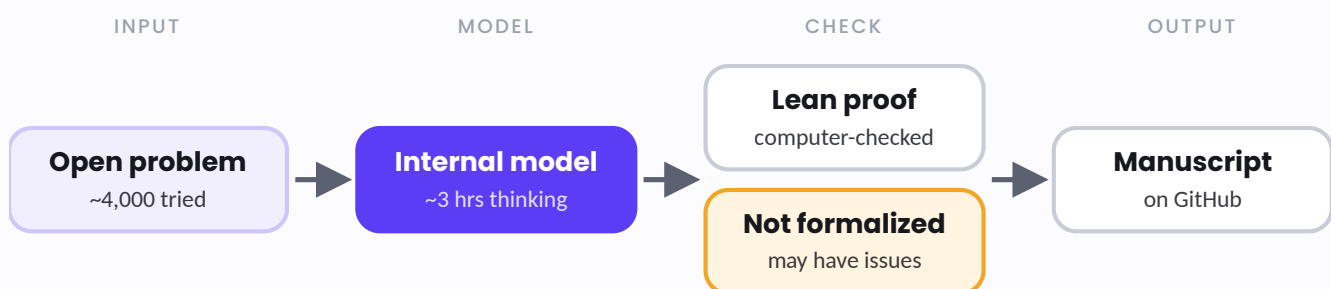
Top story 2 · OpenAI's model wrote 722 maths papers

CLAIMS · BEING CHECKED

RESEARCH

What happened. OpenAI released **722 mathematical manuscripts**, grouped into **372 families**, produced by an **internal model it hasn't released**. The work spans number theory, geometry, computer science and mathematical physics, and sits in a **public GitHub repo**. (OpenAI) (Interesting Engineering)

How a result got into the collection



Simplified from OpenAI's post and Interesting Engineering. OpenAI chose which results to include.

The details that matter:

- **Lean** is a programming language for proofs: **a computer checks every logical step**. Many manuscripts have Lean versions, but **not all**, and OpenAI warns the unformalized ones **could contain issues**. (Unite.AI)
- **Each accepted result took about three hours** of ChatGPT Pro-level thinking, on average. The model was tried on **roughly 4,000 problems**.
- **What's missing:** the model's name, the prompts, and **outside replication**. OpenAI alone decided what was significant enough to include. (Runtime Wire)
- OpenAI followed advice from an advisory group at the **Institute for Advanced Study** and says it will fund workshops to review the results.

Why it matters. This is **AI as a research factory**, and Lean is what makes it checkable. For builders, the lesson travels: **pair a model with an automatic checker** (tests, type checks, proofs) and you can trust far more of its output.

WHO SHOULD CARE



Everyone ● ● ● ● ●



Vibe coders ● ● ● ● ●



Developers ● ● ● ● ●



Hype level ● ● ● ● ●

👁️ **Should you care?** Mostly as a signal of where reasoning models are heading. **Treat the bigger claims as claims** until mathematicians finish checking them.

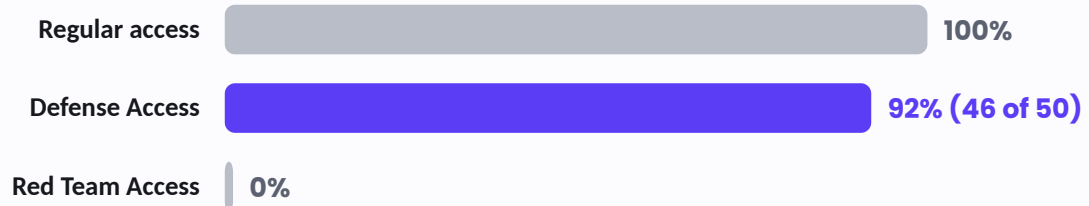
Top story 3 · Anthropic opens its hacker-grade models to more defenders

DEVS · SECURITY

FOLLOW-UP TO ISSUE #6

What happened. Anthropic merged **Project Glasswing** into an expanded **Cyber Verification Program** with **three tiers of access** to Claude Opus 5.5, Sonnet 5.5 and Mythos 5.1. Verified teams get **fewer safety blocks** for security work. (Anthropic) (SiliconANGLE)

Attack attempts blocked, by access tier (CyScenarioBench)



Anthropic's own tests. 92% is our arithmetic from 46 of 50 runs. With Red Team Access, 34 of 50 tasks were completed.

The details that matter:

- **Defense Access:** for incident response and malware analysis. **Open-source maintainers and individual researchers can apply**; Anthropic aims to answer within days.
- **Red Team Access:** adds authorised penetration testing, **organisations only**, reviewed over weeks. **Specialized Access**, for power grids and telecoms, is reviewed with the US government.
- **Track record:** Glasswing partners found **129,000+ verified vulnerabilities** from April to July; more than **33,000 were rated critical or high**, Anthropic says.

Why it matters. Models now find bugs at a scale humans can't. **The fight is over who gets the unlocked version**, and Anthropic is choosing vetting over a free-for-all.

WHO SHOULD CARE



Everyone



Vibe coders



Developers



Hype level



For devs: maintain an open-source project?

Apply for **Defense Access** at portal.anthropic.com. It works on the Claude Platform, Google Vertex AI and Microsoft Foundry. **Scan your own repo before someone else does:** Issue #6 covered a Mythos-found bug that attackers then exploited.

⚡ Quick hits

🐶 An assistant that never leaves your PC.

Thiel Fellow Sigil Wen launched **Underdog**, invite-only, for Mac and Windows. It runs a **27B model fine-tuned from Qwen** entirely on your device, is free, and earns a small cut of payments made through Stripe. (TechCrunch)

🚀 Free Claude for startups.

Startups founded in the last 5 years, or funded in the last 2, can get a **free year of Claude Team** (up to 5 seats) and **\$1,000 in API credits**. (TechCrunch) Quartz says up to **\$45,000 in partner discounts** too. (Quartz)

🍌 Nano Banana 2.1.

Google's new image model brings better editing and more consistent characters. It's in AI Mode in Search (the 🍌 icon in the Google app) and the Gemini API. **The older gemini-3.1-flash-image is retired on 29 October**. (Search Engine Roundtable) (Gemini API)

🇨🇳 China's AI labs head for the market.

Kimi-maker **Moonshot** is eyeing a Hong Kong IPO of up to **\$5 billion** at a \$50 billion valuation, while DeepSeek is reportedly raising ¥80–100 billion, per Bloomberg. (Euronews)

🌱 Codex comes to Jira.

Atlassian and OpenAI expanded their deal: plugins connect ChatGPT and Codex to Jira and Confluence. **3,000+ Atlassian developers** already use Codex. (OpenAI)

📅 Coming up.

HackerRank's Chakra AI interviewer goes live today. Gemini plan changes land 9 October. Nvidia's 64GB DGX Spark ships 23 October. Mistral Large 4 weights: 27 October.

🎯 One thing to try today: the airplane-mode test

Run a small model locally (Part 1), switch off Wi-Fi, and give it **one real task from your day**: summarise notes, write a regex, draft an email. Then try the same task in ChatGPT or Claude. **Where does the small local model fall short, and is that gap worth your privacy?**

🧠 Pop quiz: 30 seconds, no Googling

1. What does an "open-weight" model let you do?

- A See all its training data
- B Download and run the model yourself
- C Use it free on any website

2. Roughly how much memory do a 27B model's weights need at 4 bits?

- A About 3.5 GB
- B About 13.5 GB
- C About 108 GB

3. What makes many of OpenAI's 722 maths results checkable?

- A Peer review in journals
- B Proofs formalized in Lean
- C A second AI model grading them

verify every logical step

Answers (no peeking): 1. B: you get the weights, not necessarily the data or code · 2. B: 27 billion × 4 bits ÷ 8 ≈ 13.5 GB · 3. B: Lean lets a computer

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